

QED-CIS in `cqed-scf`

Formalism, working equations, and the matrix-free algorithm

Abstract

This note documents the cavity QED configuration-interaction-singles (QED-CIS) implementation in the `cqed-scf` package. It fixes the Pauli–Fierz Hamiltonian and coherent-state convention, defines the composite electron–photon basis and the index ordering used internally, gives every matrix-element block explicitly, and specifies the matrix-free σ -build and iterative eigensolver that replace the explicit Hamiltonian construction. Sections marked *implementation note* record choices specific to this code that a reader comparing against the literature would otherwise find surprising — in particular the analytic cancellation of two terms that appear explicitly in most published formulations (Sec. 2.4), and the fact that the eigenvalues are referenced to the CQED-SCF energy rather than to the correlated ground state (Sec. 6).

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1 Scope and notation

The implementation targets *Hermitian* QED-CIS with a real cavity frequency. The non-Hermitian, lossy-cavity generalisation of Ref. [3] is deliberately out of scope; the solver interface is left open to it but no complex- ω path is implemented.

symbol	meaning
i, j, k / a, b, c / p, q, r, s	occupied / virtual / general molecular orbitals
$n_o, n_v, n_{ov} = n_o n_v$	counts of occupied, virtual, and singly-excited configurations
n, m	photon-number (Fock) indices, $0 \leq n \leq N_{\text{ph}}$
$\boldsymbol{\lambda}$	cavity coupling vector, $\boldsymbol{\lambda} = \sqrt{1/\epsilon_0 V} \hat{\mathbf{e}}$
ω	cavity mode frequency (real)
$\hat{\boldsymbol{\mu}}$	total dipole operator; $\hat{\boldsymbol{\mu}}_{\text{el}}$ its electronic part
$\hat{d}$	$\boldsymbol{\lambda} \cdot \hat{\boldsymbol{\mu}}_{\text{el}}$, a one-electron operator; d_{pq} its MO matrix elements
$\langle d \rangle$	$\boldsymbol{\lambda} \cdot \langle \hat{\boldsymbol{\mu}}_{\text{el}} \rangle$ over the CQED-SCF reference
$\hat{b}^\dagger, \hat{b}$	photon creation / annihilation operators
$ \Phi_0\rangle$	CQED-SCF (or canonical HF) reference determinant
$ \Phi_i^a\rangle$	singlet spin-adapted single excitation
F_{pq}	Pauli–Fierz Fock matrix in the chosen MO basis
$(pq rs)$	two-electron repulsion integrals, chemists’ notation
g	$\sqrt{\omega/2}$

Table 1: Notation used throughout.

Only closed-shell, singlet-adapted excitations are treated. Triplets decouple from the bilinear light-matter term (their transition dipole $\langle \Phi_0 | \hat{d} | \Phi_i^a \rangle$ vanishes) but *not* from the $\mathbf{G}$ block of Sec. 4.4; they are not implemented.

2 The Pauli–Fierz Hamiltonian and the coherent-state basis

In the dipole (long-wavelength) approximation and length gauge, a single cavity mode coupled to a molecule is described by the Pauli–Fierz Hamiltonian [1, 2]

$$\hat{H}_{\text{PF}} = \hat{H}_{\text{el}} + \omega \hat{b}^\dagger \hat{b} - \sqrt{\frac{\omega}{2}} (\boldsymbol{\lambda} \cdot \hat{\boldsymbol{\mu}}) (\hat{b}^\dagger + \hat{b}) + \frac{1}{2} (\boldsymbol{\lambda} \cdot \hat{\boldsymbol{\mu}})^2, \quad (1)$$

where $\hat{H}_{\text{el}}$ is the usual electronic Hamiltonian, the third term is the bilinear light-matter coupling, and the fourth is the dipole self-energy (DSE).

2.1 Coherent-state transformation

Working directly in the photon-number basis makes the reference determinant couple to the one-photon state, because $\langle \Phi_0 | \boldsymbol{\lambda} \cdot \hat{\boldsymbol{\mu}} | \Phi_0 \rangle \neq 0$. Applying the coherent-state displacement operator that removes the reference expectation value [1, 3] replaces $\boldsymbol{\lambda} \cdot \hat{\boldsymbol{\mu}} \rightarrow \boldsymbol{\lambda} \cdot \hat{\boldsymbol{\mu}} - \langle \boldsymbol{\lambda} \cdot \hat{\boldsymbol{\mu}} \rangle$ in both the bilinear and DSE terms:

$$\hat{H}_{\text{PF}}^{\text{CS}} = \hat{H}_{\text{el}} + \omega \hat{b}^\dagger \hat{b} - \sqrt{\frac{\omega}{2}} \Delta \hat{d} (\hat{b}^\dagger + \hat{b}) + \frac{1}{2} (\Delta \hat{d})^2, \quad \Delta \hat{d} \equiv \boldsymbol{\lambda} \cdot \hat{\boldsymbol{\mu}} - \langle \boldsymbol{\lambda} \cdot \hat{\boldsymbol{\mu}} \rangle. \quad (2)$$

This is the form used throughout. Its immediate consequence, used repeatedly below, is

$$\langle \Phi_0, n | \hat{H}_{\text{PF}}^{\text{CS}} | \Phi_0, n \pm 1 \rangle = 0, \quad (3)$$

i.e. the reference determinant does not couple to its own photon replicas.

2.2 One- and two-electron decomposition of the DSE

Write $\hat{d} = \boldsymbol{\lambda} \cdot \hat{\boldsymbol{\mu}}_{\text{el}}$. Squaring the electronic dipole separates into a one-electron and a two-electron piece,

$$\frac{1}{2} \left(\sum_i \hat{d}(i) \right)^2 = \underbrace{\frac{1}{2} \sum_i \hat{d}(i)^2}_{\text{one-electron}} + \underbrace{\frac{1}{2} \sum_{i \neq j} \hat{d}(i) \hat{d}(j)}_{\text{two-electron}}. \quad (4)$$

The one-electron piece is quadrupolar: with $\hat{d} = \boldsymbol{\lambda} \cdot \mathbf{r}$ up to sign conventions, $\hat{d}^2$ contracts the Cartesian quadrupole integrals against $\lambda_x \lambda_y$. In the code this is

$$Q_{\mu\nu}^{\text{PF}} = -\frac{1}{2} \sum_{xy} \lambda_x \lambda_y Q_{\mu\nu}^{xy}, \quad (5)$$

added to the core Hamiltonian, $\mathbf{H} = \mathbf{H}_0 + \mathbf{Q}^{\text{PF}}$.

The two-electron DSE is an ERI with a separable kernel. This is the single most useful structural observation in this document. The operator $\hat{d}(1)\hat{d}(2)$ has exactly the form of a two-electron repulsion operator with the rank-one kernel

$$(pq|rs) \longrightarrow d_{pq} d_{rs}. \quad (6)$$

Every Slater–Condon result for the ERIs therefore carries over verbatim with that substitution — which is where the DSE block Δ of Sec. 4.3 comes from, and why it costs $O(N^3)$ rather than requiring any four-index quantity (Sec. 7).

2.3 The CQED-SCF reference

The reference determinant is obtained from a self-consistent solution of the Pauli–Fierz mean-field equations, giving the Fock matrix

$$F_{\mu\nu} = H_{\mu\nu}^0 + Q_{\mu\nu}^{\text{PF}} + 2J_{\mu\nu}[D] - K_{\mu\nu}[D] - N_{\mu\nu}[D], \quad N_{\mu\nu}[D] = \sum_{\rho\sigma} d_{\mu\rho} d_{\nu\sigma} D_{\rho\sigma}, \quad (7)$$

with $\mathbf{D}$ the closed-shell density. Orbitals may alternatively be taken from a cavity-free canonical HF calculation; see Sec. 5.

2.4 Implementation note: two terms that cancel analytically

Most published formulations of the coherent-state CQED-RHF Fock operator carry two additional terms that are absent from Eq. (7): a mean-field DSE contribution $\mathbf{M}$ and a one-electron coherent-state shift $\mathbf{d}^{\text{PF}}$,

$$M_{\mu\nu} = d_{\mu\nu} \text{Tr}(\mathbf{d}\mathbf{D}), \quad d_{\mu\nu}^{\text{PF}} = (\boldsymbol{\lambda} \cdot \boldsymbol{\mu}_{\text{nuc}} - \boldsymbol{\lambda} \cdot \langle \boldsymbol{\mu} \rangle) d_{\mu\nu}. \quad (8)$$

Writing $x \equiv \boldsymbol{\lambda} \cdot \langle \boldsymbol{\mu}_{\text{el}} \rangle = 2 \text{Tr}(\mathbf{d}\mathbf{D})$ and $n_{\text{nuc}} \equiv \boldsymbol{\lambda} \cdot \boldsymbol{\mu}_{\text{nuc}}$, these are $2\mathbf{M} = x\mathbf{d}$ and $\mathbf{d}^{\text{PF}} = (n_{\text{nuc}} - x - n_{\text{nuc}})\mathbf{d} = -x\mathbf{d}$, so

$$\boxed{2\mathbf{M} + \mathbf{d}^{\text{PF}} = \mathbf{0}} \quad (9)$$

identically. The corresponding energy contributions cancel against the DSE constant d_c :

$$\text{Tr}[2\mathbf{M}\mathbf{D}] = +\frac{1}{2}x^2, \quad \text{Tr}[2\mathbf{d}^{\text{PF}}\mathbf{D}] = -x^2, \quad d_c = \frac{1}{2}n_{\text{nuc}}^2 - n_{\text{nuc}}(x + n_{\text{nuc}}) + \frac{1}{2}(x + n_{\text{nuc}})^2 = +\frac{1}{2}x^2, \quad (10)$$

whose sum vanishes. The implementation therefore omits all three, which makes its Fock matrix and total energy *identical* — not merely equivalent — to formulations that carry them. Both identities are asserted numerically in the test suite, so the simplification cannot silently rot.

3 The QED-CIS ansatz and basis ordering

3.1 Basis

The composite electron–photon basis is

$$\left\{ |\Phi_0\rangle \otimes |n\rangle, |\Phi_i^a\rangle \otimes |n\rangle \right\}, \quad n = 0, \dots, N_{\text{ph}}, \quad (11)$$

of dimension $(N_{\text{ph}} + 1)(1 + n_{\text{ov}})$. Setting $N_{\text{ph}} = 1$ recovers the QED-CIS-1 method of Ref. [3]; the implementation is general in N_{ph} at no additional cost (Sec. 3.3).

3.2 Index ordering

Configurations are ordered *photon-major*, with the reference determinant leading each photon block:

$$\boxed{\text{index}(n, p) = n(1 + n_{\text{ov}}) + p, \quad p = \begin{cases} 0, & |\Phi_0, n\rangle \\ 1 + i n_v + a, & |\Phi_i^a, n\rangle \end{cases}} \quad (12)$$

Why not photon-minor. An interleaved ordering such as $\text{index} = 2(i n_v + a) + n + 2$, used in earlier reference implementations, scatters the electronic block — which is *identical* in every photon sector — across alternating rows and columns. It cannot be sliced, cannot be reused between photon sectors, forces an explicit six-deep loop over (i, a, n, j, b, m) to construct, and hard-codes two Fock states. The two orderings are related by a permutation and have identical spectra; only one of them exposes structure a σ routine can exploit.

3.3 Block structure

In the ordering of Eq. (12) the Hamiltonian is block-tridiagonal in photon number, built from just *two* photon-independent blocks:

	$n = 0$	$n = 1$	$n = 2$
$n = 0$	$A + \Delta + 0\omega\mathbf{1}$	$\sqrt{1}g\mathbf{G}$	0
$n = 1$	$\sqrt{1}g\mathbf{G}$	$A + \Delta + 1\omega\mathbf{1}$	$\sqrt{2}g\mathbf{G}$
$n = 2$	0	$\sqrt{2}g\mathbf{G}$	$A + \Delta + 2\omega\mathbf{1}$

Only $A + \Delta$ and $\mathbf{G}$ are ever constructed; the $\sqrt{n+1}$ factors are scalars. Consequently **QED-CIS- N requires no additional code beyond QED-CIS-1**. Putting the reference determinant first within each block keeps the electronic sub-block a contiguous $n_{ov} \times n_{ov}$ slice, which can be compared directly against ordinary CIS.

4 Matrix elements

All energies below are referenced to $E_{\text{CQED-SCF}}$ (see Sec. 6).

4.1 Photon block Ω

The bare photon term $\omega\hat{b}^\dagger\hat{b}$ is diagonal:

$$\langle \Phi_0, n | \omega\hat{b}^\dagger\hat{b} | \Phi_0, m \rangle = n\omega \delta_{nm}, \quad \langle \Phi_i^a, n | \omega\hat{b}^\dagger\hat{b} | \Phi_j^b, m \rangle = n\omega \delta_{nm} \delta_{ij} \delta_{ab}. \quad (13)$$

It contributes $n\omega\mathbf{1}$ to the n -th diagonal block.

4.2 Electronic block A

With a general (not necessarily canonical) Pauli–Fierz Fock matrix,

$$A_{ia,jb} = \underbrace{\delta_{ij}F_{ab} - \delta_{ab}F_{ij}}_{\text{one-electron}} + \underbrace{2(ia|jb) - (ij|ab)}_{\text{two-electron}} \quad (14)$$

and the reference–singles coupling within a photon sector is the Brillouin element

$$\langle \Phi_0, n | \hat{H} | \Phi_i^a, n \rangle = \sqrt{2} F_{ia}. \quad (15)$$

For canonical CQED-SCF orbitals $F_{ij} = \delta_{ij}\varepsilon_i$, $F_{ab} = \delta_{ab}\varepsilon_a$ and $F_{ia} = 0$, so Eq. (14) collapses to the familiar $\delta_{ij}\delta_{ab}(\varepsilon_a - \varepsilon_i) + 2(ia|jb) - (ij|ab)$ and Eq. (15) vanishes.

4.3 Dipole self-energy block Δ

The one-electron DSE contributions of Sec. 2.2 are already inside $\mathbf{F}$: the quadrupole term $\mathbf{Q}^{\text{PF}}$ sits in the core Hamiltonian and the mean-field DSE enters through $-\mathbf{N}[\mathbf{D}]$ in Eq. (7), while the coherent-state one-electron shift cancels identically (Sec. 2.4). What remains explicitly in the CIS matrix is the two-electron DSE, obtained from $2(ia|jb) - (ij|ab)$ under the separable substitution (6):

$$\Delta_{ia,jb} = \underbrace{2 d_{ia} d_{jb}}_{\text{“Coulomb-like”}} - \underbrace{d_{ij} d_{ab}}_{\text{“exchange-like”}} \quad (16)$$

The first term is rank one in the composite index (ia) ; the second is separable into $n_o \times n_o$ and $n_v \times n_v$ factors. Neither requires a four-index object, which is what makes the DSE essentially free in the σ build.

4.4 Bilinear coupling block G

The coupling operator is $-g \Delta \hat{d} (\hat{b}^\dagger + \hat{b})$ with $g = \sqrt{\omega/2}$. Its electronic part is one-electron, so between singles

$$G_{ia,jb} = \delta_{ab} d_{ij} - \delta_{ij} d_{ab} \quad (17)$$

and between the reference and a single, $\langle \Phi_0 | (-\Delta \hat{d}) | \Phi_i^a \rangle = -\sqrt{2} d_{ia}$. Combining with the photon ladder matrix elements $\langle n \pm 1 | (\hat{b}^\dagger + \hat{b}) | n \rangle = \sqrt{n+1}$ or $\sqrt{n}$ gives

$$\langle \Phi_i^a, n+1 | \hat{H} | \Phi_j^b, n \rangle = \sqrt{n+1} g G_{ia,jb}, \quad (18)$$

$$\langle \Phi_0, n+1 | \hat{H} | \Phi_i^a, n \rangle = \langle \Phi_0, n | \hat{H} | \Phi_i^a, n+1 \rangle = -\sqrt{(n+1)} \omega d_{ia}, \quad (19)$$

the latter using $\sqrt{2} g = \sqrt{\omega/2} \cdot \sqrt{2} = \sqrt{\omega}$. Note $\mathbf{G}$ is symmetric, and by Eq. (3) there is no reference–reference coupling across photon sectors.

4.5 Summary

block	element	note
$\langle \Phi_0, n \hat{H} \Phi_0, n \rangle$	$n\omega$	photon ladder
$\langle \Phi_0, n \hat{H} \Phi_0, n \pm 1 \rangle$	0	coherent-state basis, Eq. (3)
$\langle \Phi_0, n \hat{H} \Phi_i^a, n \rangle$	$\sqrt{2} F_{ia}$	zero only for canonical CQED orbitals
$\langle \Phi_i^a, n \hat{H} \Phi_j^b, n \rangle$	$A_{ia,jb} + \Delta_{ia,jb} + n\omega \delta_{ij} \delta_{ab}$	Eqs. (14), (16)
$\langle \Phi_0, n+1 \hat{H} \Phi_i^a, n \rangle$	$-\sqrt{(n+1)} \omega d_{ia}$	Eq. (19)
$\langle \Phi_i^a, n+1 \hat{H} \Phi_j^b, n \rangle$	$\sqrt{n+1} g G_{ia,jb}$	Eq. (18)

Table 2: Complete set of QED-CIS matrix elements in the basis of Sec. 3. All other blocks vanish.

5 Orbital basis: canonical HF versus CQED-SCF

Because Eq. (14) is written with a general $\mathbf{F}$, either orbital set may be used. They are *not* equivalent, and the difference is physical rather than an error:

- **Electronic non-invariance.** CIS is invariant under separate occupied–occupied and virtual–virtual rotations, but *not* under occupied–virtual rotations. Canonical HF and CQED-SCF orbitals differ by exactly such a rotation — that rotation *is* the cavity-induced orbital relaxation — so $\{|\Phi_0\rangle\} \cup \{|\Phi_i^a\rangle\}$ spans a different space in each basis. This never vanishes for CIS; it would vanish for FCI, or for CAS within its active space.
- **Photonic non-invariance.** The coherent-state displacement mixes photon-number states through an infinite expansion, so truncating at N_{ph} in the coherent-state basis is not the same truncation as at N_{ph} in the number basis. This one *does* vanish as $N_{\text{ph}} \rightarrow \infty$.

The two effects are separable by choosing the orbital basis and the photon basis independently, which makes the mixed combinations diagnostically useful: fixing CQED-SCF orbitals and varying the photon basis isolates the convergent photon-truncation error, while fixing the coherent-state basis and varying the orbitals isolates the persistent orbital-relaxation error — a quantity that measures what singles cannot recover. Note that neither basis gives a variational bound relative to the other: the two CI spaces are not nested, so no ordering of their ground-state energies is guaranteed *a priori*.

6 Energy zero and excitation energies

The Hamiltonian of Table 2 is referenced to $E_{\text{CQED-SCF}}$, i.e. $\langle \Phi_0, 0 | \hat{H} | \Phi_0, 0 \rangle = 0$. **Its lowest eigenvalue is not zero when $\lambda \neq 0$:** the bilinear term couples $|\Phi_0, 0\rangle$ to $|\Phi_i^a, 1\rangle$ and relaxes the ground state below the mean-field reference. Absolute energies and physical excitation energies are therefore

$$E_k = E_{\text{CQED-SCF}} + \epsilon_k, \quad \omega_k = E_k - E_0 = \epsilon_k - \epsilon_0, \quad (20)$$

where ϵ_k are the eigenvalues. Reporting ϵ_k directly as an excitation energy understates it by $|\epsilon_0|$. For MgH^+ at $R = 2.2 \text{ \AA}$ in cc-pVDZ with $\hbar\omega = 4.75 \text{ eV}$ and $\lambda = (0, 0, 0.0125)$ this shift is $2.51 \times 10^{-4} E_h$ — small, but larger than any sensible convergence threshold, and enough to look like a bug when two codes with different conventions are compared.

7 The matrix-free σ build

7.1 Amplitude container

A trial vector is stored as $\mathbf{X}$ of shape $(N_{\text{ph}} + 1) \times (1 + n_{\text{ov}})$, with c_n the amplitude on $|\Phi_0, n\rangle$ and $\mathbf{t}_n \in \mathbb{R}^{n_o \times n_v}$ the singles amplitudes in photon sector n . The eigensolver sees a flat vector; the σ routine sees the structured one.

7.2 Working equations

With $\mathbf{d}_{oo}, \mathbf{d}_{ov}, \mathbf{d}_{vv}$ and $\mathbf{F}_{oo}, \mathbf{F}_{ov}, \mathbf{F}_{vv}$ the MO blocks, the action $\sigma = \hat{H}\mathbf{X}$ is

$$\begin{aligned} \sigma_n^t = & \underbrace{\mathbf{t}_n \mathbf{F}_{vv} - \mathbf{F}_{oo} \mathbf{t}_n}_{\text{one-electron}} + \underbrace{\sigma^{\text{ERI}}[\mathbf{t}_n]}_{\text{two-electron}} + \underbrace{2 \mathbf{d}_{ov} \langle \mathbf{d}_{ov}, \mathbf{t}_n \rangle - \mathbf{d}_{oo} \mathbf{t}_n \mathbf{d}_{vv}}_{\text{DSE, } \Delta} + n\omega \mathbf{t}_n + \sqrt{2} c_n \mathbf{F}_{ov} \\ & + g \left[\sqrt{n} (\mathbf{d}_{oo} \mathbf{t}_{n-1} - \mathbf{t}_{n-1} \mathbf{d}_{vv}) + \sqrt{n+1} (\mathbf{d}_{oo} \mathbf{t}_{n+1} - \mathbf{t}_{n+1} \mathbf{d}_{vv}) \right] \\ & - \sqrt{n\omega} c_{n-1} \mathbf{d}_{ov} - \sqrt{(n+1)\omega} c_{n+1} \mathbf{d}_{ov}, \end{aligned} \quad (21)$$

$$\sigma_n^c = n\omega c_n + \sqrt{2} \langle \mathbf{F}_{ov}, \mathbf{t}_n \rangle - \sqrt{n\omega} \langle \mathbf{d}_{ov}, \mathbf{t}_{n-1} \rangle - \sqrt{(n+1)\omega} \langle \mathbf{d}_{ov}, \mathbf{t}_{n+1} \rangle, \quad (22)$$

where $\langle \mathbf{u}, \mathbf{v} \rangle = \sum_{ia} u_{ia} v_{ia}$ and terms with out-of-range photon indices are omitted.

7.3 The ERI action

Only σ^{ERI} requires two-electron integrals. It is evaluated in the AO basis through a generalised J/K build on the one-sided response density

$$\mathbf{D}^X = \mathbf{C}_o \mathbf{t} \mathbf{C}_v^\top, \quad \sigma^{\text{ERI}} = \mathbf{C}_o^\top \left(2\mathbf{J}[\mathbf{D}^X] - \alpha \mathbf{K}[\mathbf{D}^X] \right) \mathbf{C}_v, \quad (23)$$

queued as $\mathbf{C}_{\text{left}} = \mathbf{C}_o$ and $\mathbf{C}_{\text{right}} = \mathbf{C}_v \mathbf{t}^\top$. The density is non-symmetric, so a J/K engine accepting independent left and right coefficients is required. The scale factor α is unity for Hartree–Fock and becomes the hybrid exchange fraction for a Kohn–Sham reference.

Batching. Every trial vector and every photon sector is queued into a *single* J/K evaluation, so one eigensolver iteration costs one pass over the integrals irrespective of the number of trial vectors or the value of N_{ph} .

Explicit integrals are never formed. Eq. (23) needs only J and K in the AO basis, so the $O(N^4)$ molecular-orbital transformation is not required on the iterative path. It is performed lazily, and only for the explicit-Hamiltonian builder used as a test oracle.

7.4 Kohn–Sham references

For a Kohn–Sham reference the two-electron block additionally carries the exchange–correlation kernel, which — unlike J and K — has no molecular-orbital integral form. Rather than reconstruct its factors, the implementation calls the routine that Psi4’s own TDA engine uses,

```
wfn.twoel_Hx_full(vectors, False, "S0", singlet),
```

and combines the result exactly as Psi4 does: the returned J -like quantity already contains the XC kernel, and the K -like quantity already contains the hybrid exchange fraction and any range-separated contribution. Inheriting the convention is safer than reproducing it, because the factors then track whatever functional is in use.

A sign trap. Psi4’s `_combine_A` returns $-\mathbf{A}\mathbf{X}$; its caller negates the result. Transcribing `_combine_A` alone — even verbatim, and even though its own docstring reads $\mathbf{A}\mathbf{X} = [(\varepsilon_a - \varepsilon_i) + 2J - K]\mathbf{X}$ — yields a global sign error on the two-electron block. It is worth having a test that compares the Kohn–Sham route against the plain J/K route on a Hartree–Fock reference, where the two must agree exactly: that comparison isolates the combination and its sign from the XC kernel itself.

Consequence for the explicit build. Because no molecular-orbital form exists, the $n_{ov} \times n_{ov}$ two-electron block of a Kohn–Sham calculation is assembled by applying the action to unit vectors — one batched pass. The resulting transpose convention should be pinned with a deliberately *non-symmetric* probe; a symmetric one cannot distinguish $\mathbf{M}$ from $\mathbf{M}^\top$ and would pass with the convention inverted.

7.5 Cost

contribution	cost	note
Fock, Eq. (21)	$O(n_\sigma^2 n_v + n_o n_v^2)$	small dense products
DSE Δ	$O(n_\sigma^2 n_v + n_o n_v^2)$	no J/K build at all
coupling $\mathbf{G}$, ladder terms	$O(n_\sigma^2 n_v + n_o n_v^2)$	
ERI, Eq. (23)	one J/K pass per iteration	all vectors and sectors batched
explicit build (for comparison)	$O(n_{ov}^2)$ storage, $O(n_{ov}^3)$ diagonalisation	retained as a test oracle

Table 3: Cost of one σ evaluation. Every cavity-specific term is $O(N^3)$ or cheaper, so the cavity is essentially free relative to ordinary CIS.

8 Iterative eigensolver

8.1 Algorithm

A block Davidson–Liu procedure [6, 7] is used. The problem is real symmetric by construction, so no non-Hermitian machinery is required.

Algorithm 1 Block Davidson–Liu for QED-CIS

Require: batched action $\sigma(\cdot)$, approximate diagonal $\mathbf{h}$, guess $\mathbf{V}$, root count k

- 1: orthonormalise $\mathbf{V}$ (double Gram–Schmidt, drop near-dependent vectors)
 - 2: **repeat**
 - 3: $\mathbf{W} \leftarrow \sigma(\mathbf{V}_{\text{new}})$ ▷ one batched pass
 - 4: $\mathbf{S} \leftarrow \mathbf{V}\mathbf{W}^\top$; symmetrise
 - 5: $(\theta, \mathbf{y}) \leftarrow \text{eigh}(\mathbf{S})$, keep the lowest k
 - 6: $\mathbf{X} \leftarrow \mathbf{y}^\top \mathbf{V}$ ▷ Ritz vectors
 - 7: $\mathbf{R} \leftarrow \mathbf{y}^\top \mathbf{W} - \theta \mathbf{X}$ ▷ residuals
 - 8: **if** $\max_k \|\mathbf{R}_k\| < \tau$ **then converged**
 - 9: **end if**
 - 10: **if** $\dim \mathbf{V} + k > s_{\text{max}}$ **then**
 - 11: **thick restart:** $\mathbf{V} \leftarrow \mathbf{Y}^\top \mathbf{V}$, $\mathbf{W} \leftarrow \mathbf{Y}^\top \mathbf{W}$ keeping $\geq 2k$ Ritz vectors
 - 12: **end if**
 - 13: $\delta_k \leftarrow \mathbf{R}_k / (\theta_k - \mathbf{h})$ ▷ diagonal preconditioner
 - 14: **normalise** each δ_k , then orthogonalise against $\mathbf{V}$ and drop dependents
 - 15: **until** converged or iteration limit
-

8.2 Preconditioner and guess

The diagonal used for preconditioning is

$$h_{(n,ia)} = F_{aa} - F_{ii} + n\omega + 2d_{ia}^2 - d_{ii}d_{aa} \left[+ 2(ia|ia) - (ii|aa) \right], \quad h_{(n,0)} = n\omega, \quad (24)$$

where the bracketed two-electron terms are included only when the integral engine can supply them cheaply; the preconditioner need not be exact. Guess vectors are unit vectors on the lowest diagonal entries, augmented with the photonic references $|\Phi_0, n\rangle$ for all n .

8.3 Practical notes

These three points are recorded because each was found by failure rather than by design, and each is easy to get wrong:

1. **Normalise preconditioned corrections before testing for linear dependence.** The drop test must measure how much of a direction is *new*, not the magnitude of the correction. Preconditioned residuals shrink as the solver converges, so testing raw norms causes the solver to mistake a small correction for a dependent one and stagnate — eigenvalues converged to 10^{-13} while residuals plateau near 10^{-6} .
2. **Restart thick.** Collapsing to exactly k Ritz vectors discards nearly-converged directions just outside the requested set and the iteration thrashes. Retaining $\gtrsim 2k$ vectors fixes it. A maximum subspace too small to expand after a restart cannot converge and should be clamped upward rather than obeyed.
3. **Degeneracies are the normal case.** At $\lambda = 0$ every root is $(N_{\text{ph}} + 1)$ -fold degenerate, with molecular point-group degeneracies on top. The block size should be at least the expected degeneracy, orthogonalisation should be double Gram–Schmidt with an explicit drop threshold, and converged roots must be returned with their full multiplicity. Deduplicating a computed spectrum is never correct: it cannot distinguish a basis replication from a physically degenerate state.

Convergence thresholds should be tight. Rabi splittings scale with $|\lambda|$ and can be of order $10^{-4} E_h$, so residual tolerances of 10^{-6} to 10^{-8} are appropriate rather than the 10^{-4} typical of TDDFT drivers.

9 Properties

9.1 Polaritonic character

For root k with amplitudes $(c_n^{(k)}, \mathbf{t}_n^{(k)})$, the photon number and sector weights are

$$\langle \hat{b}^\dagger \hat{b} \rangle_k = \sum_n n w_n^{(k)}, \quad w_n^{(k)} = (c_n^{(k)})^2 + \|\mathbf{t}_n^{(k)}\|^2, \quad \sum_n w_n^{(k)} = 1. \quad (25)$$

These are what distinguish a lower from an upper polariton and a polariton from a dark state; without them a list of eigenvalues cannot be assigned.

9.2 Transition dipoles and oscillator strengths

The electronic dipole is diagonal in photon number, so

$$\langle \Psi_0 | \hat{\boldsymbol{\mu}} | \Psi_k \rangle = \sum_n \left[\sqrt{2} (c_n^{(0)} \langle \boldsymbol{\mu}_{ov}, \mathbf{t}_n^{(k)} \rangle + c_n^{(k)} \langle \boldsymbol{\mu}_{ov}, \mathbf{t}_n^{(0)} \rangle) + \langle \mathbf{t}_n^{(0)}, \mathbf{t}_n^{(k)} \rangle \boldsymbol{\mu}_{vv} - \boldsymbol{\mu}_{oo} \mathbf{t}_n^{(k)} \rangle \right], \quad (26)$$

and $f_{0k} = \frac{2}{3} \omega_k |\langle \Psi_0 | \hat{\boldsymbol{\mu}} | \Psi_k \rangle|^2$. Any constant (reference-expectation) part of the dipole drops out for $k \neq 0$ by orthogonality. The singles–singles term matters here in a way it does not for ordinary CIS, because the polaritonic ground state carries genuine singles amplitudes.

Caution: intensities are basis-dependent under exact degeneracy. A numerical diagonalisation returns an arbitrary orthogonal mixture within a degenerate manifold, and oscillator strength redistributes freely among its members — one member can carry all the intensity and its partner none. Only the *sum over a complete degenerate manifold* is well defined. This is acute at $\boldsymbol{\lambda} = \mathbf{0}$, $\omega = 0$, where even the ground state is an arbitrary mixture of $|\Phi_0, 0\rangle$ and $|\Phi_0, 1\rangle$. Excitation energies are unaffected, being equal across a manifold by definition.

10 Limiting cases and verification

The following limits are exact and make useful regression tests:

limit	expected result
$\boldsymbol{\lambda} = \mathbf{0}, \omega = 0$	ordinary singlet CIS, each root $(N_{\text{ph}} + 1)$ -fold degenerate
$\boldsymbol{\lambda} = \mathbf{0}, \omega \neq 0$	CIS spectrum replicated at every $n\omega$
any ordering permutation	identical spectrum (Sec. 3)
$\boldsymbol{\sigma}$ applied to all unit vectors	reproduces the explicit Hamiltonian elementwise
$N_{\text{ph}} \rightarrow \infty$	coherent-state and number-basis results converge to each other

Table 4: Exact limits used for verification.

The $\boldsymbol{\sigma}$ column test deserves emphasis: applying the matrix-free action to every unit vector and comparing the assembled matrix elementwise against an explicit build validates every sign, factor and index at once, and is far more diagnostic than comparing converged eigenvalues.

11 Relationship to linear-response QED-TDDFT

The method described here is a *configuration-interaction* theory in a composite electron–photon Fock basis. It is not the linear-response QED-TDDFT of Yang *et al.* [4], and the two differ by more than a choice of approximation. Because a Kohn–Sham reference (Sec. 7.4) makes the present method superficially resemble their TDA-PF model, the differences are worth setting out explicitly.

11.1 Their ansatz is a product, and their photon is a displacement

The QED-TDDFT working equations of Ref. [4] follow from a time-dependent variational treatment of the *product* wavefunction $|\Phi(t)\rangle |\chi(t)\rangle$, in which $|\Phi(t)\rangle = e^{-\hat{A}_e} |\Phi_0\rangle$ carries orbital rotations only and $|\chi(t)\rangle = e^{-\hat{A}_{\text{ph}}} |\chi_0\rangle$ is a *coherent-state displacement* of the photon, $\hat{A}_{\text{ph}} = \sum_{\alpha} (-C_{\alpha}^* \hat{b}_{\alpha} + C_{\alpha} \hat{b}_{\alpha}^{\dagger})$. Linearising with $C_{\alpha}(t) = M_{\alpha} e^{-i\Omega t} + N_{\alpha} e^{+i\Omega t}$ makes M and N the two Fourier components of that displacement amplitude — not amplitudes on a photon-number state. Their Appendix A reaches

the same place from the other direction: the photon obeys a driven harmonic-oscillator equation of motion, with $\delta q_\alpha = \sqrt{\hbar/2\omega_\alpha}(M_\alpha + N_\alpha)$.

Their TDA-PF model is therefore

$$\begin{bmatrix} \mathbf{A} + \mathbf{\Delta}' & \hbar\mathbf{g}^\dagger & \hbar\tilde{\mathbf{g}}^\dagger \\ \hbar\mathbf{g} & \hbar\omega & 0 \\ \hbar\tilde{\mathbf{g}} & 0 & \hbar\omega \end{bmatrix} \begin{bmatrix} \mathbf{X} \\ M \\ N \end{bmatrix} = \hbar\Omega \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} \mathbf{X} \\ M \\ N \end{bmatrix}, \quad (27)$$

over a space of dimension $n_{ov} + 2$: the electronic single excitations plus two photonic amplitudes.

11.2 What follows

No $|\Phi_i^a, 1\rangle$ configurations. A product ansatz cannot represent “electron excited *and* photon in $|1\rangle$ ” independently of $|\Phi_0\rangle \otimes |1\rangle$, so those configurations are absent from Eq. (27) — and with them the matrix element $\langle\Phi_0, 0|\hat{H}_{\text{PF}}|\Phi_i^a, 1\rangle$, which in the present method is $-\sqrt{\omega}d_{ia}$ (Eq. (19)), and the entire $\langle\Phi_i^a, 0|\hat{H}|\Phi_j^b, 1\rangle = gG_{ia,jb}$ block carrying the excited-state dipole differences d_{ij} and d_{ab} .

Counter-rotating terms are still present, differently. It would be wrong to read the above as “TDA-PF makes the rotating-wave approximation.” The counter-rotating physics enters Eq. (27) through the $\hbar\tilde{\mathbf{g}}^\dagger$ coupling of $\mathbf{X}$ to the negative-metric amplitude N ; deleting that coupling is precisely what defines their TDA-RWA model. The distinction is between an implicit, RPA-like treatment and an explicit configuration.

The ground state. Ω in Eq. (27) is an excitation energy measured from an unrelaxed reference. The present method diagonalises $\hat{H}_{\text{PF}}$ over a basis that includes $|\Phi_0, 0\rangle$, so its lowest root *is* a correlated ground state, relaxed below the mean-field reference by the coupling to $|\Phi_i^a, 1\rangle$ (Sec. 6).

The dipole self-energy is treated at different levels. Their

$$\Delta'_{ai,bj} = \sum_{\alpha} \lambda_{ai}^{\alpha} \lambda_{bj}^{\alpha} \quad (28)$$

is rank one: it comes from evaluating $\langle(\boldsymbol{\lambda} \cdot \Delta\hat{\boldsymbol{\mu}})^2\rangle$ over a single determinant as the square of a one-body expectation value, i.e. a Hartree-like mean-field DSE. The present method treats the DSE as a genuine two-electron operator with the separable kernel of Eq. (6), giving both terms of Eq. (16), $2d_{ia}d_{jb} - d_{ij}d_{ab}$. The exchange-like term has no counterpart in $\mathbf{\Delta}'$ and is not recoverable by rescaling.

Metric. Eq. (27) carries the indefinite metric $\text{diag}(1, 1, -1)$ and is a pseudo-Hermitian generalised eigenvalue problem; only their RWA and Jaynes–Cummings models reduce to a standard Hermitian problem. The present method has an identity metric throughout.

Neither treatment is a special case of the other, and the naming should reflect that: with a Kohn–Sham reference this method is QED-CIS built on QED-Kohn–Sham orbitals, not a Tamm–Dancoff approximation to QED-TDDFT.

12 Not included

For completeness, the following are deliberately absent and would each require new derivation rather than new code paths: triplet excitations; non-Hermitian (complex- ω , lossy-cavity) treatments as in

	TDA-PF, Ref. [4]	this work
photon degree of freedom	coherent-state displacement	Fock states $0..N_{\text{ph}}$
electron–photon ansatz	product	configuration interaction
$ \Phi_i^a, n \geq 1\rangle$ configurations	absent	present
counter-rotating terms	implicit, via N	explicit matrix elements
ground state	not relaxed	correlated, variational
DSE	mean-field, rank one	two-electron, Eq. (16)
metric	diag(1, 1, -1)	identity
photon truncation	one amplitude per mode	arbitrary N_{ph}

Table 5: The two theories are related in spirit and different in content. The electronic block $\mathbf{A}$ and the bilinear coupling $\mathbf{g}$ agree between them up to spin-adaptation conventions; nothing else does.

Ref. [3]; multi-mode cavities; and the linear-response QED-TDDFT family of Refs. [4, 5], which as Sec. 11 explains is a distinct theory rather than an extension of this one.

An open question worth recording: what the correct $\mathbf{B}$ -block analogue is for a Fock-basis method of this kind — that is, an RPA-like generalisation in which de-excitation amplitudes are defined over the composite electron–photon basis rather than over a photonic displacement. That is not the same object as the $\mathbf{B} + \mathbf{\Delta}$ block of Eq. (8) of Ref. [4].

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